MODULE 05

Big Data and Data Engineering

A concise 10-page starter workbook for Data Science students

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Learning goal

This workbook introduces the core ideas in big data and data engineering, with a topic-by-topic study path, practical checkpoints, and reflection prompts. Use it as a starting point and supplement it with your class notes and recommended references.

Note: These are sample learning materials created for this demo portal, not official university course notes.

Data Sources and Collection

Concept overview

In this section, focus on the key principles behind **data sources and collection**. Start by defining the problem, identifying the data or assumptions involved, and explaining why the concept matters in a real data science workflow. Keep a short glossary of unfamiliar terms as you study.

A practical way to learn

1. Write the idea in your own words.
2. Work through one small example by hand.
3. Try the same example with a real or synthetic dataset.
4. Record what changed when you adjusted an assumption or parameter.

Checkpoint

Explain data sources and collection to a classmate in three sentences. Then list one situation where it would be useful, one limitation to remember, and one question you still have.

Study notes	
Key term	My explanation
Example	What did I observe?
Open question	What should I investigate next?

Relational Databases and SQL

Concept overview

In this section, focus on the key principles behind **relational databases and sql**. Start by defining the problem, identifying the data or assumptions involved, and explaining why the concept matters in a real data science workflow. Keep a short glossary of unfamiliar terms as you study.

A practical way to learn

1. Write the idea in your own words.
2. Work through one small example by hand.
3. Try the same example with a real or synthetic dataset.
4. Record what changed when you adjusted an assumption or parameter.

Checkpoint

Explain relational databases and sql to a classmate in three sentences. Then list one situation where it would be useful, one limitation to remember, and one question you still have.

Study notes	
Key term	My explanation
Example	What did I observe?
Open question	What should I investigate next?

ETL and ELT Pipelines

Concept overview

In this section, focus on the key principles behind **etl and elt pipelines**. Start by defining the problem, identifying the data or assumptions involved, and explaining why the concept matters in a real data science workflow. Keep a short glossary of unfamiliar terms as you study.

A practical way to learn

1. Write the idea in your own words.
2. Work through one small example by hand.
3. Try the same example with a real or synthetic dataset.
4. Record what changed when you adjusted an assumption or parameter.

Checkpoint

Explain etl and elt pipelines to a classmate in three sentences. Then list one situation where it would be useful, one limitation to remember, and one question you still have.

Study notes	
Key term	My explanation
Example	What did I observe?
Open question	What should I investigate next?

Data Warehousing and Data Lakes

Concept overview

In this section, focus on the key principles behind **data warehousing and data lakes**. Start by defining the problem, identifying the data or assumptions involved, and explaining why the concept matters in a real data science workflow. Keep a short glossary of unfamiliar terms as you study.

A practical way to learn

1. Write the idea in your own words.
2. Work through one small example by hand.
3. Try the same example with a real or synthetic dataset.
4. Record what changed when you adjusted an assumption or parameter.

Checkpoint

Explain data warehousing and data lakes to a classmate in three sentences. Then list one situation where it would be useful, one limitation to remember, and one question you still have.

Study notes	
Key term	My explanation
Example	What did I observe?
Open question	What should I investigate next?

Distributed Computing Concepts

Concept overview

In this section, focus on the key principles behind **distributed computing concepts**. Start by defining the problem, identifying the data or assumptions involved, and explaining why the concept matters in a real data science workflow. Keep a short glossary of unfamiliar terms as you study.

A practical way to learn

1. Write the idea in your own words.
2. Work through one small example by hand.
3. Try the same example with a real or synthetic dataset.
4. Record what changed when you adjusted an assumption or parameter.

Checkpoint

Explain distributed computing concepts to a classmate in three sentences. Then list one situation where it would be useful, one limitation to remember, and one question you still have.

Study notes	
Key term	My explanation
Example	What did I observe?
Open question	What should I investigate next?

Apache Spark Fundamentals

Concept overview

In this section, focus on the key principles behind **apache spark fundamentals**. Start by defining the problem, identifying the data or assumptions involved, and explaining why the concept matters in a real data science workflow. Keep a short glossary of unfamiliar terms as you study.

A practical way to learn

1. Write the idea in your own words.
2. Work through one small example by hand.
3. Try the same example with a real or synthetic dataset.
4. Record what changed when you adjusted an assumption or parameter.

Checkpoint

Explain apache spark fundamentals to a classmate in three sentences. Then list one situation where it would be useful, one limitation to remember, and one question you still have.

Study notes	
Key term	My explanation
Example	What did I observe?
Open question	What should I investigate next?

Data Quality and Governance

Concept overview

In this section, focus on the key principles behind **data quality and governance**. Start by defining the problem, identifying the data or assumptions involved, and explaining why the concept matters in a real data science workflow. Keep a short glossary of unfamiliar terms as you study.

A practical way to learn

1. Write the idea in your own words.
2. Work through one small example by hand.
3. Try the same example with a real or synthetic dataset.
4. Record what changed when you adjusted an assumption or parameter.

Checkpoint

Explain data quality and governance to a classmate in three sentences. Then list one situation where it would be useful, one limitation to remember, and one question you still have.

Study notes	
Key term	My explanation
Example	What did I observe?
Open question	What should I investigate next?

Cloud Data Platforms

Concept overview

In this section, focus on the key principles behind **cloud data platforms**. Start by defining the problem, identifying the data or assumptions involved, and explaining why the concept matters in a real data science workflow. Keep a short glossary of unfamiliar terms as you study.

A practical way to learn

1. Write the idea in your own words.
2. Work through one small example by hand.
3. Try the same example with a real or synthetic dataset.
4. Record what changed when you adjusted an assumption or parameter.

Checkpoint

Explain cloud data platforms to a classmate in three sentences. Then list one situation where it would be useful, one limitation to remember, and one question you still have.

Study notes	
Key term	My explanation
Example	What did I observe?
Open question	What should I investigate next?

Mini Project: Design a Data Pipeline

Concept overview

In this section, focus on the key principles behind **mini project: design a data pipeline**. Start by defining the problem, identifying the data or assumptions involved, and explaining why the concept matters in a real data science workflow. Keep a short glossary of unfamiliar terms as you study.

A practical way to learn

1. Write the idea in your own words.
2. Work through one small example by hand.
3. Try the same example with a real or synthetic dataset.
4. Record what changed when you adjusted an assumption or parameter.

Checkpoint

Explain mini project: design a data pipeline to a classmate in three sentences. Then list one situation where it would be useful, one limitation to remember, and one question you still have.

Study notes	
Key term	My explanation
Example	What did I observe?
Open question	What should I investigate next?